

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION
FIRST SEMESTER 2026-2027
Course Handout Part II

Date: 03/08/2026

In addition to part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : CS F320
Course Title : Foundations of Data Science
Instructor-in-charge : NAVNEET GOYAL (goel@)
Course TA : Soumya R Saju (p20240432@)

Catalog Description

Data Science is the study of the generalizable extraction of knowledge from data. Unprecedented advances in digital technology during the second half of the 20th century and the data explosion that ensued in the 21st century is transforming the way we do science, social science, and engineering. Application of data science cut across all verticals. A data scientist requires an integrated skill set spanning mathematics, probability and statistics, optimization, and branches of computer science like databases, machine learning etc. The course aims at providing the mathematical and computer science foundations required for data science.

Text & Reference Books:

- T1. Foundations of Data Science - Avrim Blum, John Hopcroft, Ravi Kannan, January 2018.
- R1. Machine Learning: An Algorithmic Perspective – Stephen Marsland, CRC Press, 2e, 2015.
- R2. Pattern Recognition & Machine Learning – Christopher M Bishop, Springer, 2006.
- R3. An Introduction to Data Science – Jeffrey Saltz and Jeffrey Stanton, Sage Publications, September 2017.

LECTURE PLAN (42)

Topic	Topic Details	No. of Lectures	Chapter Reference
Course Overview & Introduction to Data Science	1. Motivation/course objectives 2. Some motivating applications 3. Data Dimensionality 4. Types of Data	3	T1 – Ch. 1 Class Notes + web resources
High-dimensional data & Curse of Dimensionality	1. Characteristics of High-dimensional data a. Law of large numbers b. Geometry of HD c. Vectors in HD d. Gaussians in HD 2. Curse of Dimensionality (CoD) a. Sparsity b. Nearest Neighbor search c. Concentration of Lp-norm 3. Dimensionality Reduction Techniques – PCA & SVD	4	T1 – Chs. 2, 3

Frequentist vs. Bayesian approach to Probability	1. Frequentist Approach 2. Bayesian Approach 3. Prior to Posterior – Bayes’ Theorem 4. MLE vs. MAP	2	Class Notes + https://sites.google.com/site/bayestutorial/
Probability Distributions and Mixture Models	1. Exponential family of distributions (Bernoulli, Beta, Binomial, Dirichlet, Gamma, & Gaussian) 2. Mixture Models – Mixture of Gaussians	2	R2 – Ch.2, Appendix B
Optimization Techniques	1. Unconstrained & Constrained Optimization 2. Convex Optimization & Lagrange Multipliers 3. Quadratic Programming 4. Primal/dual 5. Kernels 6. Gradient Descent & its variants	3	Class Notes
Function Approximation Techniques	1. Basis Functions 2. Splines 3. Mixture Models	2	R1: Chs. 5,7
Tensors	1. Introduction to Tensors 2. Tensor Algebra 3. Tensor Calculus 4. Modeling multidimensional data using Tensors	3	http://web.iitd.ac.in/~pmvs/courses/mcl702/tensors.pdf
Machine Learning Basics	1. Supervised Learning a. Regression (polynomial, linear basis function models, ridge regression, Lasso, SVM), Evaluation Metrics (MAE, RMSE, R^2) b. Classification (Naive Bayes, Decision Tree, Logistic regression, K-NN), Evaluation metrics (Precision, Recall, F1-score, ROC, AUC, Cross-validation) 2. Unsupervised Learning a. K-means Clustering b. Expectation Maximization Clustering c. Self-Organizing Maps (SOMs) 3. Model Underfitting & Overfitting - Regularization 4. Model Selection & Complexity a. Occam’s Razor b. VC dimension c. Structural Risk Minimization d. Bias-Variance Decomposition	8	T1 – Chs. 5,7 R1 – Chs. 2-4,8,12,14 R2 – Chs. 1,3,9 R3 – Ch. 18
Probabilistic Graphical Models	1. Markov Models 2. Hidden Markov Models (HMM) 3. Markov Random Field (MRF)	3	T1: Ch. 9 R1: Ch. 16 R2: Ch. 8
Time-series Data & Analytics	1. Importance & Characteristics of time series data 2. Sources of time series data 3. Similarity Metrics & Dynamic Time Warping 4. Multi-variate Time Series Data – IoT Data 5. Time Series analytics:	5	Class Notes + web resources

	a. Regression b. Classification c. Clustering d. Anomaly Detection		
Data Visualization	1. Basics of Data Visualization 2. Multivariate data visualization	2	Class Notes + web resources
Introduction to Databases	1. Graph Databases 2. Vector Databases	2	Class Notes + web resources
		Total:42	

Evaluation Scheme:

Component	Duration	Weightage	Date (Time)
Midsem Test (Closed Book)	90 Mins.	30% (60 marks)	09/10 FN1
Assignments (02)*	Take Home	Non-evaluatory	
Evaluatory Labs. (best 08 out of 10)		20% (40 marks)	
Quiz	45 Mins.	10% (20 marks)	25/09 F9 (Class Hour)
Comprehensive Exam (partly open)	120 Mins.	40% (80 marks)	13/12 FN

** First Assignment will be uploaded in the first week of September and the second assignment immediately after Midsem. Assignment-based questions will be there in midsem (up to 5%/10 marks - based on assignment 1) and Comprehensive examination (up to 10%/20 marks - based on assignment 2).*

Labs. using Python: Structured lab. sessions as per the schedule give in the timetable.

Notices: All notices will be uploaded on NALANDA only.

Chamber Consultation Hours: Navneet Goyal (Th 10)

Makeup Policy: To be granted only in case of serious illness or emergency.

Plagiarism Policy: If any student is found involved in any kind of plagiarism in any of the evaluation components, the matter will be directly reported to the Examination Committee.

NC Policy: Students securing 10% or less marks will get an NC grade. Students in the [10-15%] bracket are also likely to get NC.

Instructor-in-charge
CS F320